

11

Society, Law and Ethics

11.1 INTRODUCTION

The digital world is changing at a tremendous pace. New communication technologies have opened up new possibilities. However, by using them you also expose yourself, and others, to various risks. Many people have trouble assessing these risks, especially with regard to the subject of safe digital communication. This is particularly true of people working in regimes with high levels of censorship. Your data can be used or misused by others as well—governments, companies, or even other individuals, sometimes even unintentionally.

Information Technology (IT) plays a central role in commerce, industry, government, education, entertainment and society at large. Its economic and social benefits hardly need



explanation. But, like any other technology, IT also has adverse implications and some negative impacts on society. It poses and creates problems related to ethics and, in general, contains three main types of ethical issues: personal privacy, access right and harmful actions. Let us look closely at these issues, exploring in each case the ways in which they affect public reactions to this technological change.

How to protect yourself, your sources, or your friends? What are the safe routes to take? How do you secure your personal data? This chapter aims to address these issues to help you choose your own 'level' of safety. We will also learn about the laws which can provide us with complete safety while working online. This chapter also discusses computer ethics or standards of conduct pertaining to computers. It also deals with how computing professionals should make decisions regarding professional and social conduct.

Basically, computer ethics describe the difference between ethical and unethical. For example, while it is easy to duplicate copyrighted electronic (or digital) content, computer ethics would suggest that it is wrong to do so without the author's approval. And while it may be possible to access someone's personal information on a computer system, computer ethics would advise that such an action is unethical.

Let us discuss these topics in detail.

11.2 DIGITAL FOOTPRINTS

This is the term used to describe the trail, traces or "footprints" that people leave online while using smartphones, tablets, computers, etc. Digital footprints are recorded with each and every online activity that we perform—be it interactions on the social media, chatting and connecting with friends on the social media sites, online shopping, or locations through Facebook check-ins, etc.



These digital footprints can be compared to footprints that you leave behind when walking along a beach. Every step that you take leaves an impression on the sandy surface that allows another person to see and follow your marks. Digital footprints are also termed as Digital Tattoos. However, a beach footprint gets washed away after some time whereas digital footprints stay forever and cannot be done away with.

Our digital footprints can be created and used with or without our knowledge. They include websites we visit, emails we send, and any information we share online, etc., along with the computer's IP address, location, and other device-specific details. Such data could be used for targeted advertisements or could also be misused or exploited. Thus, it is always good to be aware of the data trail we might be leaving behind. This awareness should make us cautious about what we write, upload or download or even browse online.

Digital footprints are classified into two types:

1. Active digital footprints
2. Passive digital footprints

An active digital footprint is created when a user intentionally shares their personal information either through social media platforms or websites and apps. Examples of active digital footprints are:

- (a) Sharing of personal information on Facebook, Instagram, Twitter and other social media platforms.
- (b) Working with online forms, such as signing up while logging on to mail accounts to send or receive emails or text messages.
- (c) Accidental/intentional acceptance to install cookies on users' respective devices when prompted by the browser.

A passive digital footprint is created when information is collected from the user without their knowledge. Examples of passive digital footprints are:

- (a) Websites that install cookies in a user's device without disclosing it.
- (b) Apps and websites that use geo-location to detect a user's location.
- (c) Social media news channels and advertisers that use comments, likes and shares of the user to know about their profiles and areas of interest, based on which they send advertisements as per the user's requirements and interests.

Besides browser, most of our digital footprints are stored in servers where the applications are hosted. We may not have access to remove or erase that data, neither do we have any control on how that data will be used. Therefore, once a data trail is generated, even if we later try to erase data about our online activities, digital footprints still remain. There is no guarantee that digital footprints will be fully eliminated from the internet. Therefore, we need to be more cautious while being online! All our online activities leave a data trace on the internet as well as on the computing device that we use. This can be used to trace the user, their location, device and other usage details.

11.3 DIGITAL SOCIETY AND NETIZEN

Today, we are living in a digital society in which every aspect of our lives is being profoundly affected by the digitalisation of data. We perform most of our activities like shopping, education, social networking, entertainment, health care, paying our bills, etc., using these technologies. 'Netizen' itself is a contraction of the words internet and citizen and refers to both a person who uses the internet to participate in society and an individual who has accepted the responsibility of using the internet in productive and socially responsible ways. Being a responsible netizen, we must respect anyone using internet. A responsible netizen must abide by net etiquettes, communication etiquettes and social media etiquettes.

11.3.1 Netiquettes

Netiquette is the short form of "internet etiquette" or communication etiquettes over the internet. Just like etiquette—a code of polite behaviour in society—netiquette is a code of good behaviour while working on the internet. It includes several aspects of the internet, such as social media, email, online chat, web forums, website comments, multiplayer gaming and other types of online communication.



While working online, netiquettes revolve around the general idea of respecting others online. Given below are the Do's and Don'ts that one should follow while working on the internet.

Do's	Don'ts
• Keep messages and posts brief. • Reread your posts or emails to make sure they say what you intend. • Remember that you leave a digital footprint. So, be careful what you write. • Use discretion. • Include a subject line in an email. • Protect personal information. • Obey copyright laws. • Stay focused and stick to the topic.	• Don't post inflammatory/offensive comments. • Don't write in ALL CAPS. It is considered as SHOUTING on the net. • Do not respond to internet trolls/personal attacks. • Avoid posting private or embarrassing images/comments. • Don't use sarcasm. It might be misinterpreted. • Do not violate copyright laws. Make sure your work is your own or properly cited. • Do not exclude people or talk behind their backs. • Don't spam others by sending large number of unsolicited emails.

11.3.2 Communication Etiquettes

Digital communication refers to the rules or etiquettes to be followed while sending email, texting, instant messaging, talking on the cell phone, audio or video conferencing, posting on forums, social networking sites, etc. All these are great ways to connect with people in order to exchange ideas, share data and knowledge. Good communication over email, chat room and other such forums requires a digital citizen to abide by the communication etiquettes as follows.



1. **Be Precise:** It means that whenever any communication is required, we should be precise. Respect the time and avoid some useless messages to forward or reply. Do not expect an instant reply. While sending attachments, we should avoid a large file. If it is required to be sent then send the link of cloud storage like Google Drive, OneDrive or Dropbox, etc.
2. **Be Polite:** In communication, we should always be polite whether we agree to something or not. We should reply politely without any aggression or being abusive.
3. **Be Credible:** While making a comment on a forum, we should sound credible. Also, while making a comment on somebody, we should always try to go through their previous comments and judge their credibility before typing a comment.

11.3.3 Social Media Etiquettes

In the current digital era, we are familiar with different kinds of social media platforms and we may have an account on Facebook, Google+, Twitter, Instagram, Pinterest, or YouTube channel. Social media are websites or applications that enable users to participate in social networking by creating and sharing content with others in the community. These platforms encourage users to share their thoughts and experiences through posts or pictures. In this way users can interact with other online users of those social media apps or channels. This is why the impact and outreach of social media have grown exponentially. It has begun to shape the outcome of politics, business, culture, education and more. In social media too, there are certain etiquettes that we need to follow such as:



(a) Be Secure

- **Choose password wisely:** News of breach or leakage of user data from social network often attracts headlines. Users should be wary of such possibilities and must know how to safeguard themselves and their accounts. The minimum one can do is to have a strong password and change it frequently, and never share personal credentials like username and password with others.
- **Know who you befriend:** Social networking sites usually encourage connecting with users (making friends), sometime even those we don't know or have never met. However, we need to be careful while befriending unknown people as their intentions could possibly be malicious.

- *Rewrite of fake information*: Fake news, messages and posts are common in social networks. As a user, we should be aware of them and be able to figure out whether a news, message or post is genuine or fake. Thus, we should not blindly believe in everything that we come across on such platforms. We should apply our knowledge and experience to validate such news, message or post.

(b) Be Reliable

- *Think before uploading*: We can upload almost anything on social network. However, remember that once uploaded, it is always there in the remote server even if we delete the files. Hence, we need to be cautious while uploading or sending sensitive or confidential files which have a bearing on our privacy.

11.4 DATA PROTECTION

Data protection addresses the practices, safeguards, and binding rules put in place to protect our personal information and ensure that it remains safe while working online. In short, we should be able to decide what information should be shared, who has access to it, for how long and for what reason, and whether rights to modify some of this information should be given or not.



Data protection secures sensitive data—from databases to big data, cloud, file system and other crucial applications. Elements of data that can cause substantial harm, embarrassment, inconvenience and unfairness to an individual, if breached or compromised, is called sensitive data. Examples of sensitive data include biometric information, health information, financial information or other personal documents, images or audios or videos. Privacy of sensitive data can be implemented by encryption, authentication and other secure methods to ensure that such data is accessible only to the authorised user and is for a legitimate purpose.

One must ensure that there shouldn't be any kind of unprotected data or data breach as it may contribute to several serious issues:

1. Physical data loss
2. Attack by a virus, worm or malicious code
3. Targeted by hackers
4. DDoS(Distributed denial of service) attacks
5. Loss of money
6. Intellectual property at risk

Thus, it is mandatory to protect data. This can be achieved by taking proper and careful measures while working online in the following ways:

1. Through Data Encryption.
2. Not sharing private information such as passwords, credit card credentials, etc.
3. Not oversharing on social networking sites using public domain.
4. Adopting complete security mechanism for protection against possible threats.
5. Avoiding opening phishing emails.
6. Being careful about using Wi-Fi connections.
7. Being alert to impersonators.

11.5 INTELLECTUAL PROPERTY RIGHTS

You must have heard the word **property**; it is generally used to mean a possession of, more specifically, something to which the owner has legal rights.

You might have also encountered the phrase **intellectual property**. This term has become more commonplace during the past few years, especially in the context of computer ethics. But what exactly does it refer to?

Intellectual property refers to creations of the intellect (hence the name); inventions, literary and artistic works, symbols, names, images and designs used in commerce are part of it.

Intellectual Property Rights (IPRs) are specific legal rights that allow their owner to completely benefit over the use of his/her creation for a certain period of time. These rights also entitle him/her to prevent others from using or dealing with his/her product without prior permission and force them to stop and compensate for any damages.

Intellectual property is usually divided into two branches, namely **industrial property** which, broadly speaking, protects inventions and **copyright**, which protects literary and artistic works.



**INTELLECTUAL
PROPERTY RIGHTS**

Fig. 11.1(a): Intellectual Property Rights

Intellectual property is divided into two categories:

(a) **Industrial property**, which includes inventions (patents), trademarks, industrial designs, commercial names, designations and geographic indications (location-specific brands), etc.

- **Patent:** A patent is usually granted for inventions. Unlike copyright, the inventor needs to apply (file) for patenting an invention. When a patent is granted, the owner gets an exclusive right to prevent others from using, selling, or distributing the protected invention. Patent gives full control to the patentee to decide whether or how the invention can be used by others. Thus, it encourages inventors to share their scientific or technological findings with others. Term of every patent in India is 20 years from the date of filing of patent application after which it can be freely used.
- **Trademark:** A trademark includes any visual symbol, word, name, design, slogan, label, etc., that distinguishes a brand or commercial enterprise from other brands or commercial enterprises. For example, no company other than Reebok can use the Reebok brand to sell shoes or clothes. It also prevents others from using a confusingly similar mark, including words or phrases, such as "Reebokk". However, it may be possible to apply for the Reebok trademark for unrelated goods like textbooks.

(b) **Copyright** includes literary and artistic works such as novels, poems, plays, films, musical works, artistic works such as drawings, paintings, photographs, sculptures, architectural designs and computer software. Copyright is a legal concept, enacted by most governments, giving the creator of the original work exclusive rights to it, usually for a limited period.



Fig. 11.1(b): Copyright

It provides exclusive publication, distribution and usage rights to the copyright holder. This means whatever content the author created cannot be used or published by anyone else without the consent of the author.

The rights include right to copy (reproduce) a work, right to create derivative works based upon it, right to distribute copies of the work to the public, and right to publicly display or perform the work. It prevents others from copying, using or selling the work.

In a nutshell, copyright laws protect intellectual property which includes literary and artistic works such as novels, poems, plays, films, musical works, artistic works such as drawings, paintings, photographs and sculptures, and architectural designs.

However, unlike protection of inventions, copyright law protects only the form of expressions of ideas, not the ideas themselves.

11.6 VIOLATION OF IPR

Intellectual property (IP) infringement refers to any violation or breach of protected intellectual property rights. IP infringement can range from piracy (unauthorized use), copy, reproduction, or distribution of materials protected by IP rights to counterfeiting which is the practice of replicating genuine goods with the intent to mislead the recipient. Violation of intellectual property right may happen in one of the following ways:

11.6.1 Plagiarism

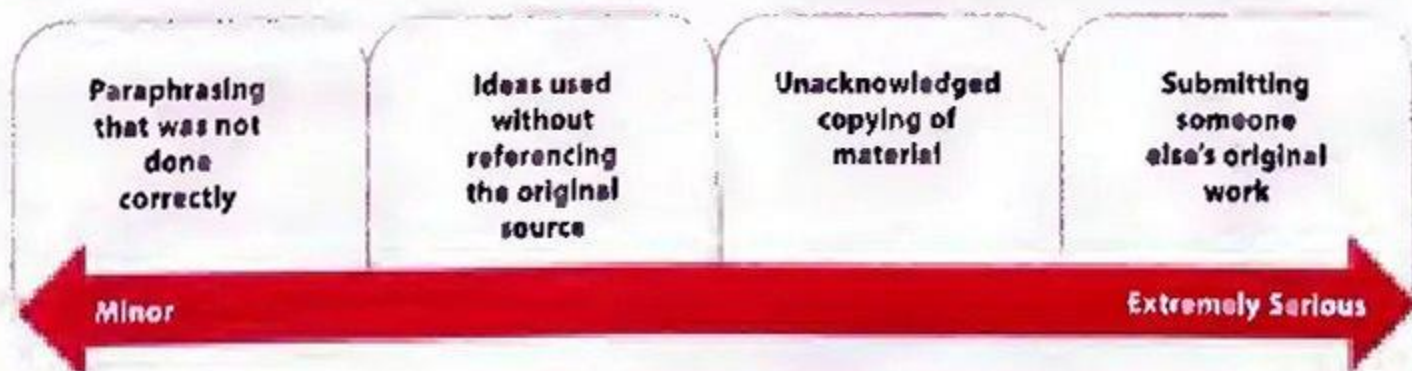
Plagiarism is copying someone else's work and then passing it off as one's own. It is morally bad because it is an act of stealing. In other words, it is copying information and not giving the author credit for it.

Copying programs written by other programmers and claiming them as your own could be an act of plagiarism. It involves lying, cheating, theft, and dishonesty.



Fig. 11.2: Plagiarism

Plagiarism can be classified as **accidental/unintentional** or **deliberate/intentional**. **Intentional plagiarism** includes copying someone else's work, cutting and pasting blocks of text or any kind of media (audio, video files or movie clips) from electronic sources without documenting and at the same time publishing it on the web without the permission of developers/creators. On the other hand, unintentional/accidental plagiarism involves careless paraphrasing (changing the words or sentence construction of a copied document), quoting text excessively along with poor documentation. Usually, accidental plagiarism cases are less serious in comparison to deliberate plagiarism that may result in serious implications.



How to avoid Plagiarism

Plagiarism is a bad practice and can be avoided by taking the following simple measures:

1. Use your own words and ideas.
2. Always provide reference or give credit to the source from where you have received your information.
3. If, by any chance, it becomes necessary to use someone's exact words, don't forget to put them in quotes and give credit using in-text citations.
4. As far as possible, try to include the source in your works citing page number.
5. Cite the name of the website, a URL, or the name of authors, and acknowledge them if you have used their work after rearranging the order of a sentence and changing some of the words.
6. Take information in the form of bulleted notes in your own words rather than copying the entire content or complete sentences.
7. Use online tools to check for plagiarism.

11.6.2 Copyright Infringement

Copyright infringement is the use or production of copyrighted material without the permission of the copyright holder. There are many types and forms of copyright infringement.

- Downloading an image from the internet and using it in your project.
- Recording a film in a movie theatre.
- Using copyrighted images on your website.
- Modifying an image and then displaying it on your company's website.
- Downloading music or films without paying for their use.
- Copying any literary or artistic work without a license or written agreement.

11.6.3 Trademark Infringement

Trademark infringement is defined as the unauthorized use of a trademark or service mark on or in connection with goods and/or services in a manner that is likely to cause confusion, deception, or mistake about the source of the goods and/or services. Trademark owners can take legal action if they believe their marks are being infringed.

11.7 DIGITAL RIGHTS MANAGEMENT

Digital Rights Management (DRM) refers to access control technologies that are used by hardware manufacturers, publishers, copyright holders and individuals to limit the use of digital content and devices. The term is used to describe any technology that inhibits the use of digital content that is not desired or intended by the content provider.

It attempts to control use of digital media by preventing access, copying or conversion to other formats by the end-users. There are several instances in our day-to-day lives when we intend to download or save some digital content from the internet but



Fig. 11.3(a): Unauthorized Access to Digital Content

are barred from doing so and usually get an error which signifies the restriction imposed on the user to access this content (Fig. 11.3(a)). It refers to a scheme that controls access to copyrighted material using technological means. In essence, DRM removes usage control from the person in possession of digital content and puts it in the hands of a computer program.

DRM technologies have enabled publishers to enforce access policies that not only disallow copyright infringements but also prevent lawful, fair use of copyrighted works, or even impose restrictions on non-copyrighted works that they distribute.

DRM is most commonly used by the entertainment industry (e.g., film and recording).

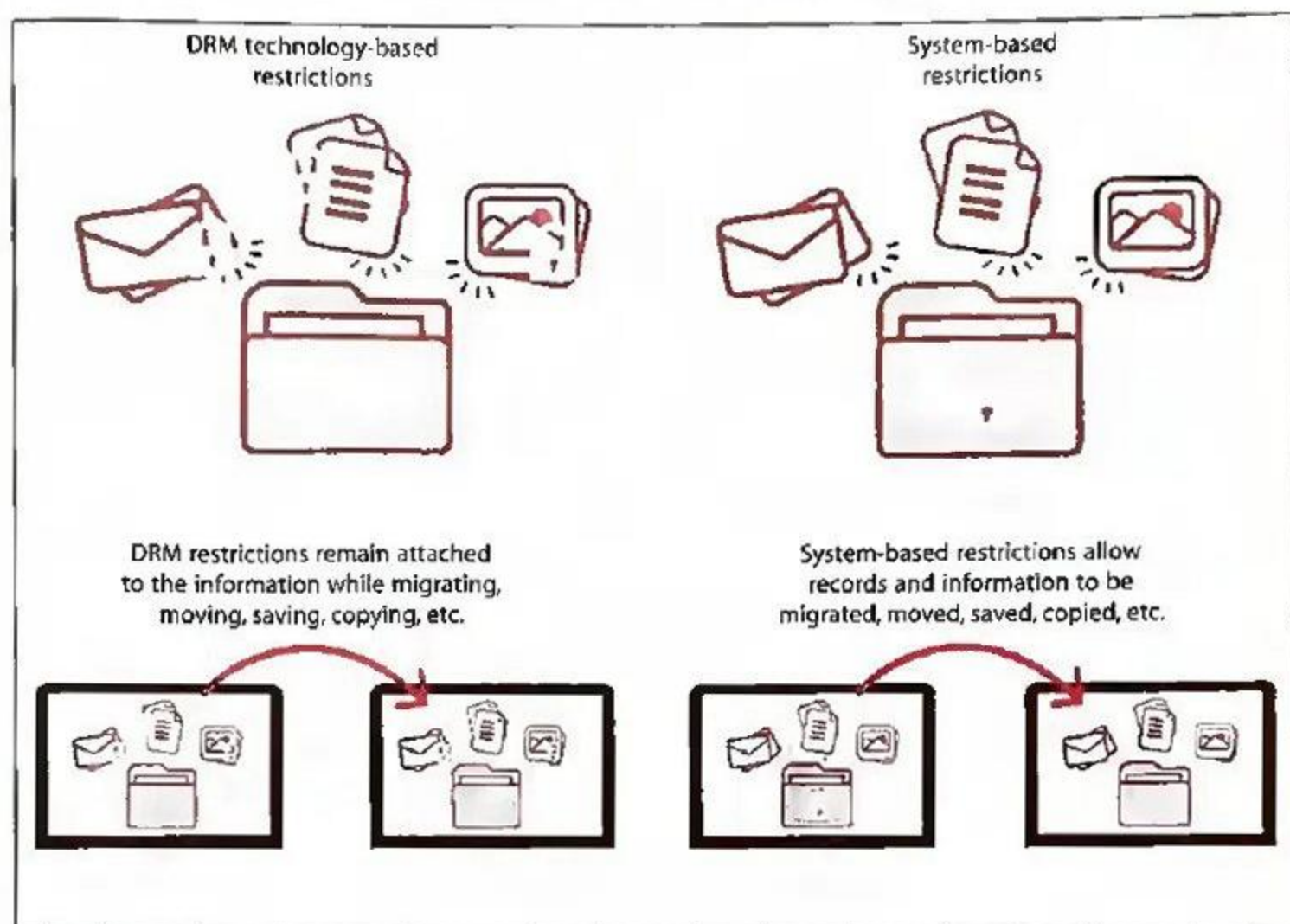


Fig. 11.3(b): Significance of Implementing DRM Technology

Many online music stores such as Apple's iTunes Store as well as many e-book publishers have implemented DRM. An organization which intends to use DRM-based data should take the following into consideration: Intellectual Property (IP) rights, Copyright, Payment mechanisms, Security, E-commerce and ERP systems, and most important of all, Licensing.

DRM provides a better mechanism to safeguard the data. DRM constraints are stronger than system-based restrictions as in the case of data copying, moving or saving. These constraints persist with the data and we can't access this data even for saving purpose. On the contrary, system-based restricted data is portable, which means it can be copied and saved to a different location other than its destination computer. Hence, DRM restrictions are far wide and more effective than system-based restrictions. Thus, these constraints contribute significantly to data privacy and security. In a nutshell, DRM protects rights of digital media producers while enabling access to fair use.

11.8 LICENSING

Software Licensing is the legal right to run or the privilege given to you by a company to access their application (or program).

A common example of licensing that we observe in our daily lives is the licence we purchase for proprietary software such as Windows OS to get them installed on our computer system as shown in Fig. 11.4(a). The given figure shows Windows 8 as a copyright of Microsoft, which indicates that this software is a licensed product of Microsoft and has to be legally purchased for use. It comes with a licensing agreement which is to be read first and to be agreed upon for the successful installation and usage of the software under the link “**Read the Microsoft Software License Terms**”.

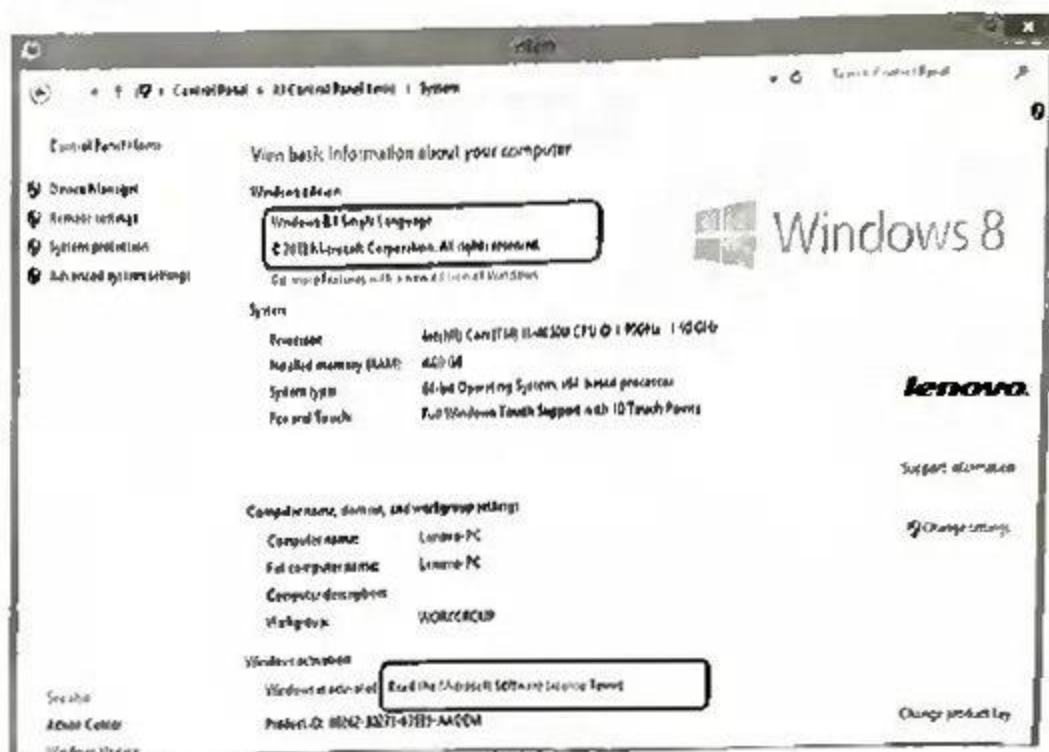


Fig. 11.4(a): Software Licensing

As soon as you click on this link, you get to see a licensing agreement (Fig. 11.4(b)) with terms and conditions. A licensing agreement governs the use of licensed software. Licensing agreements typically allow the software to run on a limited number of computers and allow copies to be made, though solely for backup purposes.

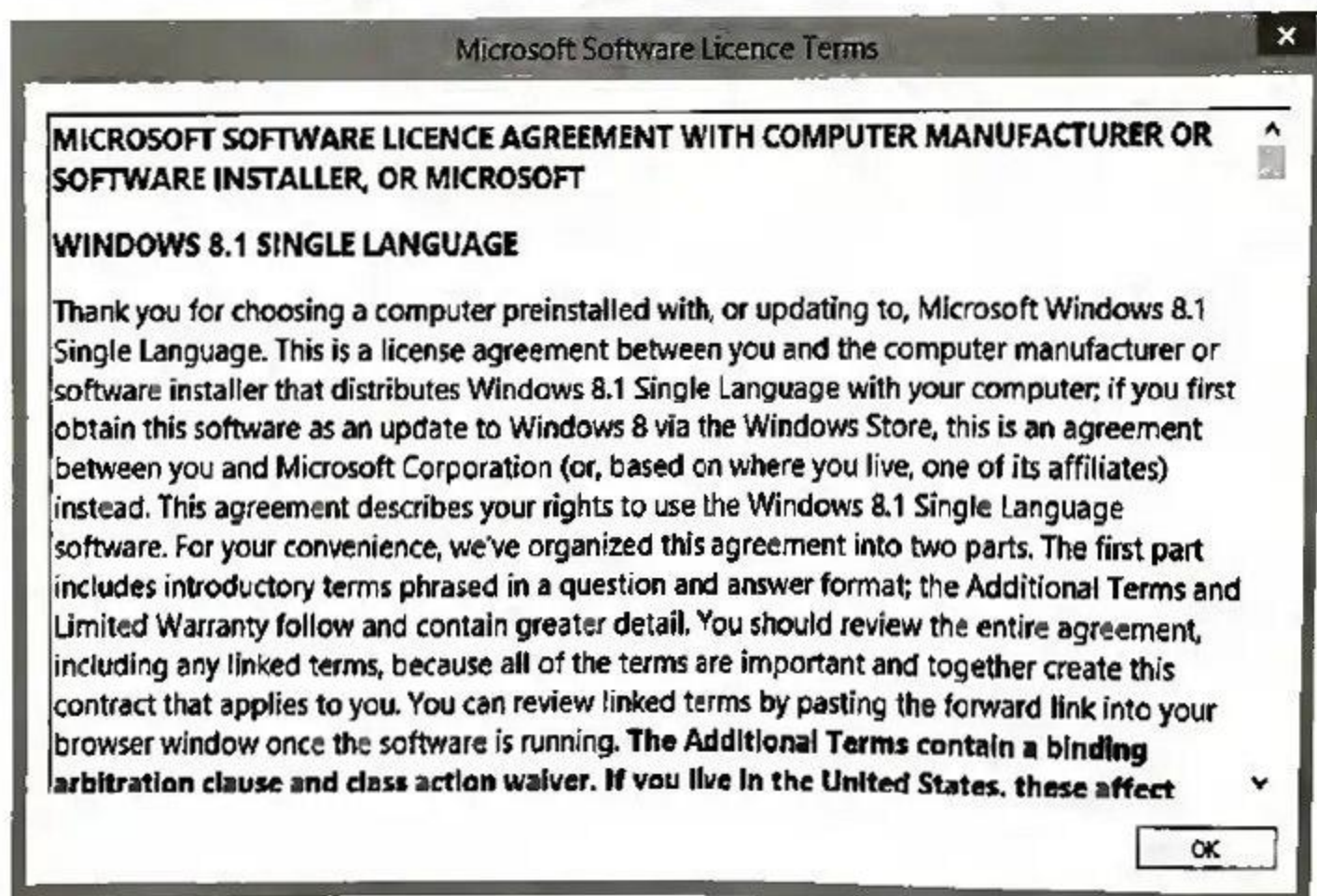


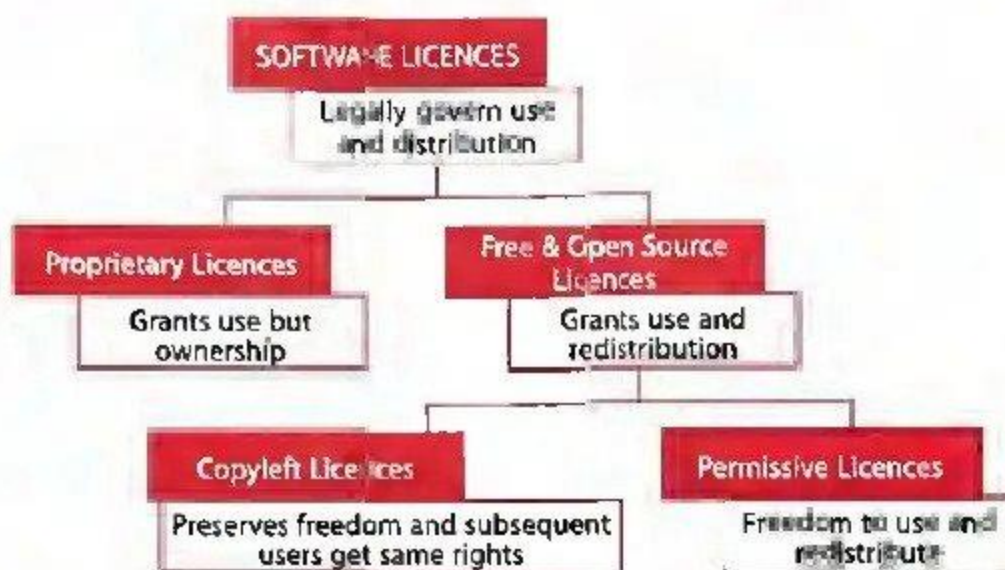
Fig. 11.4(b): A Licensing Agreement

Advantages of using Licensed Software

It is always good to use licensed software as licensing provides advantages while working with these software.

1. By using licensed software, you are able to contribute to the further development of the program you are using. It also helps the economy generate not just computer-related jobs but opportunities for other people/businesses whose work is highly dependent on "written" technology.
2. Apart from the legal obligation to use licensed software, It comes with outright support not found in "pirated" software.

Thus, licences provide rules and guidelines for others to use your work, *for example*, Open Source Licences help others to contribute to your work or project without seeking special individual permission to do so. Licences can be further classified into two types on the basis of criteria of their use—Proprietary Licences and Free and Open Source Licences.



In **Proprietary Licences**, exclusive rights in the software are retained with the owner/developer/publisher. They reserve all the freedoms and rights to use and distribute these proprietary software. Violation of the proprietary licence amounts to copyright infringement. They restrict inspection, modification of source code and further distribution.

However, freeware (*i.e.*, no-cost software) can be distributed using proprietary licences. Most commercially available software are covered under proprietary licences.

On the other hand, **Free and Open Source Licence** refers to a software that users can safely run, adapt and redistribute without legal restraint, and which emphasizes on freedom.

Open source software is a software with source code that is publicly available under a licence that gives users the right to study, change, and distribute that software and emphasizes on security, cost-saving and transparency. Hence, free and open source software (FOSS) allows using, copying, studying and modifying the software, and the source code to be openly shared so that people are encouraged to voluntarily improve the design of the software.

CTM: An OSS (open source software) refers to freedom to use, share and/or modify the source code and allow copyrights to other users.

OSS is further categorized as Permissive and Copyleft licences. Permissive licences provide a royalty-free licence to do virtually anything with the source code. They permit using, copying, modifying, merging, publishing, distributing, sublicensing and/or selling, but distribution can only be made without the source code as source code modifications can lead to permissive licence violation, e.g., Apache, W3C, etc. In case of Copyleft licences, source code has to be provided. Also, distribution and modification of source code is permitted. Examples include General Public Licence (GPL), Creative Commons Licence, Lesser General Public Licence (LGPL), Mozilla Public Licence (MPL), etc.

Let us discuss important features of OSS licences.



Free as in Freedom



Apache

GPL (General Public Licence) has the following characteristics:

- **Copying the Software:** There is no limit to copying the code. You can copy it on your own server, on your client's server, on your local workstations, anywhere and as many times as you want.
- **Distribution:** You can distribute it in your thumb or hard drives, you can distribute the code under this licence with a download link on your website, or you can print the code on paper; in fact, whatever form of distribution you prefer.
- **Charge a Fee:** You can charge someone for the software, but remember to give them a copy of GNU GPL, which would tell them that they could get the software free from elsewhere. This also gives you a chance to tell them why you are charging for it.

A CC (Creative Commons) Licence has four basic parts:

1. **Accreditation:** Author must be acknowledged as the creator of the work. The work can then be modified, distributed, copied and used otherwise.
2. **Shared with CC:** The work can be modified and distributed but only under CC Licence.
3. **Non-commercial:** Work can be modified and distributed but not for commercial purposes. The word "commercial" is a bit vague in its meaning since no solid-lined definition is available.
4. **No Derivative Works:** You can copy and distribute the licensed work but you can't modify it in any way or create work based on the origin.

APACHE Licence offers the following salient features:

1. **Rights are Never-ending:** Once the rights under Apache Licence have been granted, you can continue to use them forever. There's no need to renew them.
2. **Worldwide Authority of Rights:** Even if rights are granted to one country, automatically, they are considered as granted to all countries.
3. **Rights for No Fee or Royalty:** No charge, neither upfront nor per usage, or on any other basis is applicable.

4. **Rights are Irrevocable:** No one can ever tell you that your derivative of the code that was licensed under Apache can't be in use anymore. (A clause in the licence states that if you sue someone over patent infringement on anything under this licence, then your licence is terminated, but that only applies to patented work, and as long as you don't sue anyone over the work, you won't have to worry about it.)

1.9 OPEN SOURCE AND OPEN DATA

Open source software is a software which provides us the freedom to work, use, distribute and modify the source code. We have already discussed OSS characteristics and its examples in detail in the previous section.

There is an important term which conceptually runs parallel to open source software, which is "Open Data".

Open data is all stored data which could be made accessible in public interest without any restrictions on its usage and distribution.

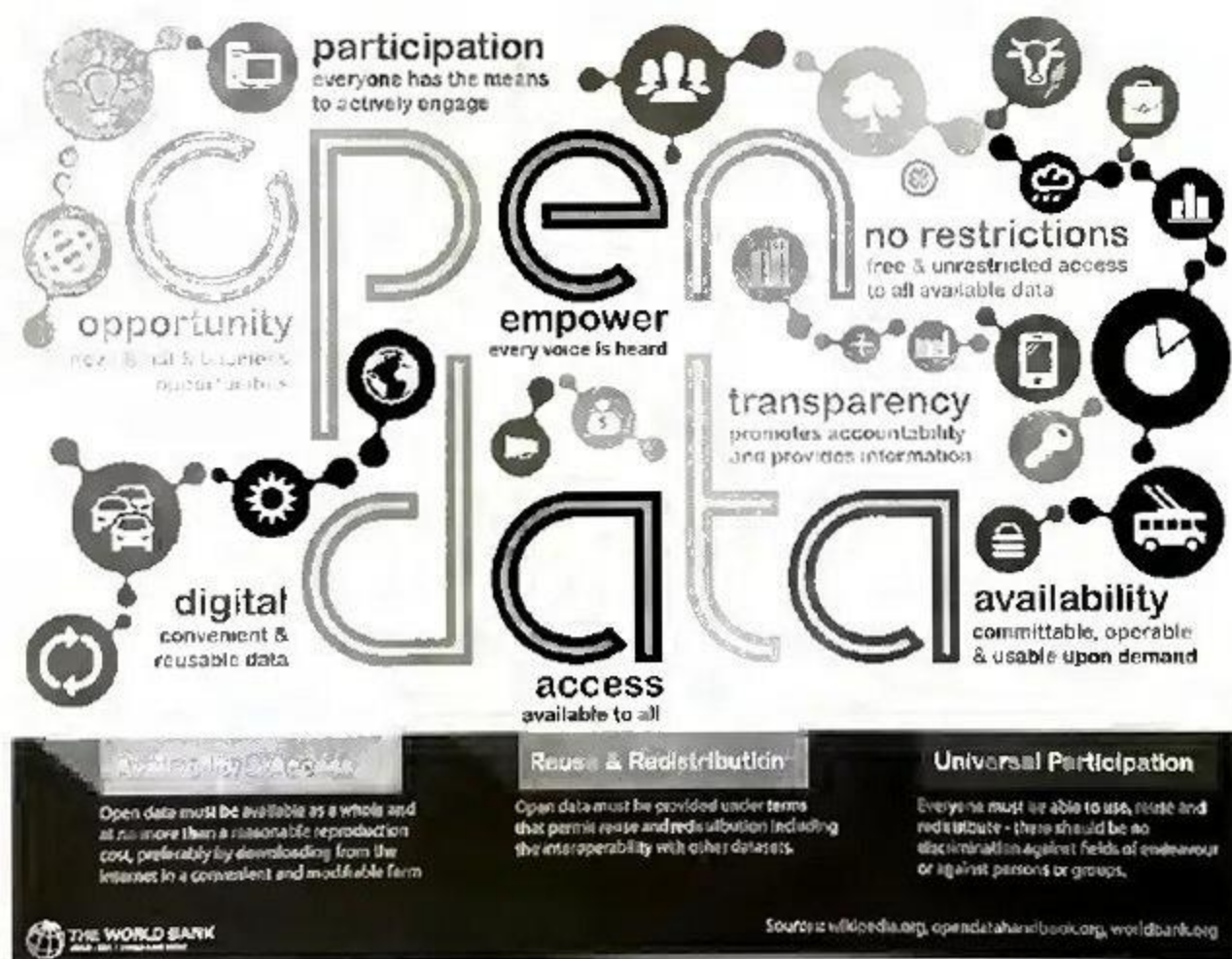


Fig. 11.5(a): Features of Open Data

- **Availability and Access:** The data must be available as a whole and at no more than a reasonable reproduction cost, preferably by downloading from the internet. The data must also be available in a convenient and modifiable form.
- **Reuse and Redistribution:** The data must be provided under terms that permit reuse and redistribution including intermixing with other datasets.

- **Universal Participation:** Everyone must be able to use, reuse and redistribute—there should be no discrimination against fields of endeavour or against persons or groups. For example, 'non-commercial' restrictions that would prevent 'commercial' use, or restrictions of use for certain purposes (e.g., only in education) are not allowed.

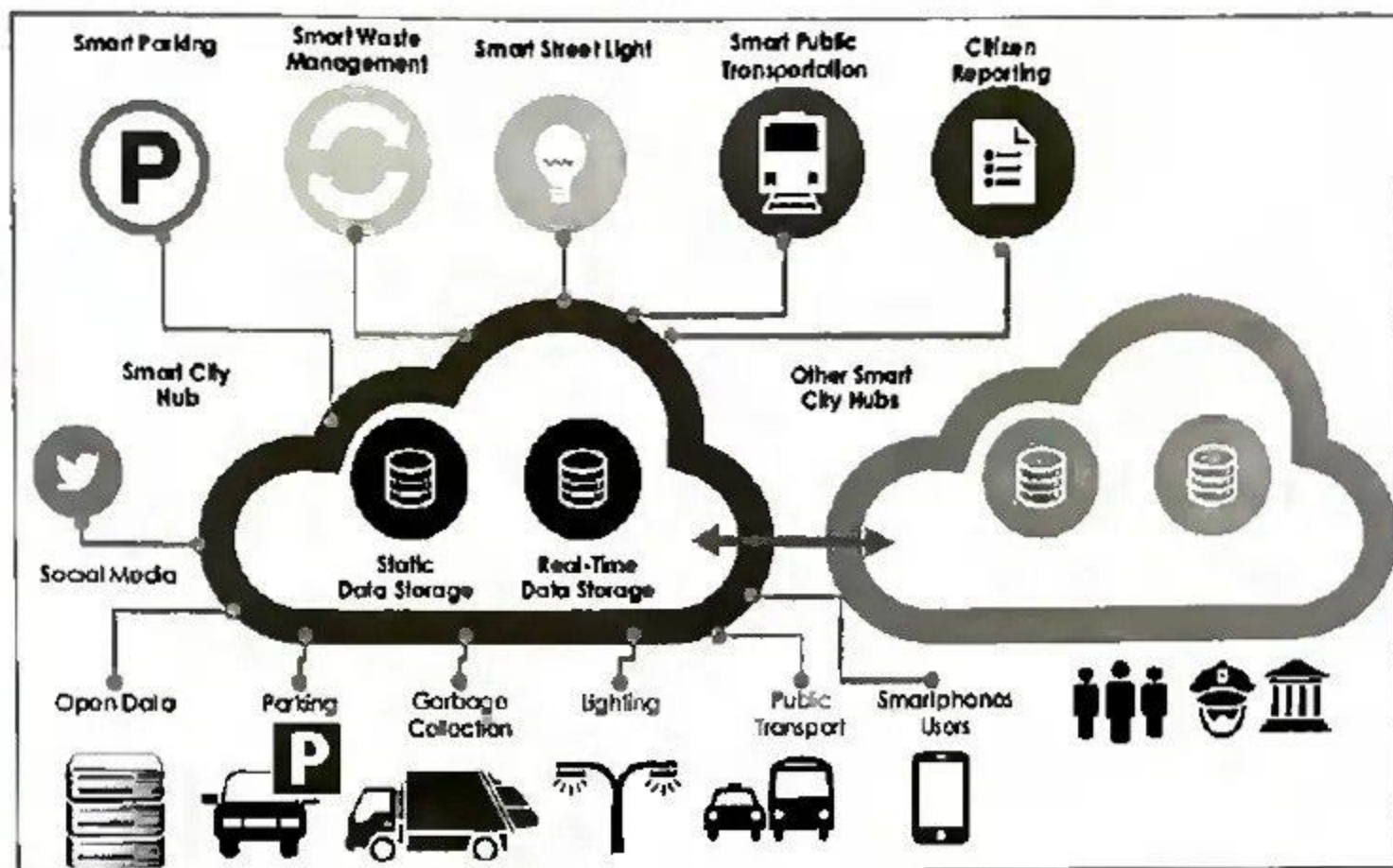


Fig. 11.5(b): Applications of Open Data

11.10 PRIVACY LAWS

Privacy refers to the ability to control what information one reveals about oneself over the internet and to control who can access that information. There are several threats to a user's privacy on the internet, which are as follows:

1. User-tracking is done through:

- Internet Service Providers (capable of observing any internet-related activity of the user)
- Cookies (parcels of text sent by a server) tracking and maintaining specific information of the user
- Data logging (may include recording times when the computer is in use or which websites are visited)
- Spyware programs
- Web bug
- Social engineering
- Phishing
- Malicious proxy server
- Search engines



Fig. 11.6: Online Privacy

- ### 2. Illegal and harmful content:
- There is a need to fight against illegal and harmful content on the internet with legal tools.

3. **Online social networks:** Risks range from identity theft to online and physical stalking, embarrassment, discrimination and blackmailing.

To provide protection from these online privacy threats, privacy laws were made which prohibit disclosure or misuse of information about private individuals.

Privacy laws refer to laws that deal with regulating, storing and using personally identifiable information about individuals, which can be collected by governments, public or private organizations or other individuals.

Privacy laws are considered within the context of an individual's privacy rights or within reasonable expectation of privacy.

The Information Technology Act, 2000 has two Sections relating to privacy:

- Section 43A, which deals with implementation of reasonable security practices for sensitive personal data or information and provides for compensation to a person affected by wrongful loss or wrongful gain.
- Section 72A, which provides for imprisonment for a period of up to 3 years and/or a fine of up to ₹ 5,00,000 to a person who causes wrongful loss or wrongful gain by disclosing personal information about another person while providing services under the terms of lawful contract.

The IT Act, 2000 has provisions that permit interception, monitoring and decryption of digital communication. It provides for collection and monitoring of traffic data. It allows the government to set a national encryption standard. Projects like the Central Monitoring System, NATGRID, Phone & Internet Interception used by police are some crucial measures taken up by the Government of India for safeguarding privacy laws. We will be discussing IT Act, 2000 in detail in successive sections.

11.11 CYBERCRIME

Cybercrime is defined as a crime in which a computer is the object of the crime (hacking, phishing, spamming) or is used as a tool to commit an offence (child pornography, hate crimes, etc.).



Cyber crimes are carried out against an individual, a group, an organization or even against a country, with the intent to directly or indirectly cause physical harm, financial loss or mental harassment. A cybercriminal attacks a computer or a network to reach other computers in order to disable or damage data or services.

Cybercriminals may use computer technology to access personal information, business trade secrets or use the internet for exploitative or malicious purposes. Criminals can also use computers for communication and document or data storage. Criminals who perform these illegal activities are often referred to as hackers.

Common types of cybercrimes include online banking information theft, identity theft, online predatory crimes and unauthorized computer access. More serious crimes like cyber terrorism are also of significant concern.

Cybercrime encompasses a wide range of activities but these are generally divided into two categories:

1. Crimes that target computer networks or devices: These types of crimes include viruses and denial-of-service (DoS) attacks.
2. Crimes that use computer networks to advance other criminal activities: These types of crimes include cyberstalking, phishing and fraud or identity theft.

11.11.1 Hacking

Hacking is an act of unauthorized access to a computer, computer network or any digital system. Hackers usually have technical expertise of the hardware and software.

They look for bugs to exploit and break into the system. Hacking, when done with a positive intent, is called ethical hacking. Such ethical hackers are known as white hat hackers. They are specialists in exploring any vulnerability or loophole during the testing of the software.

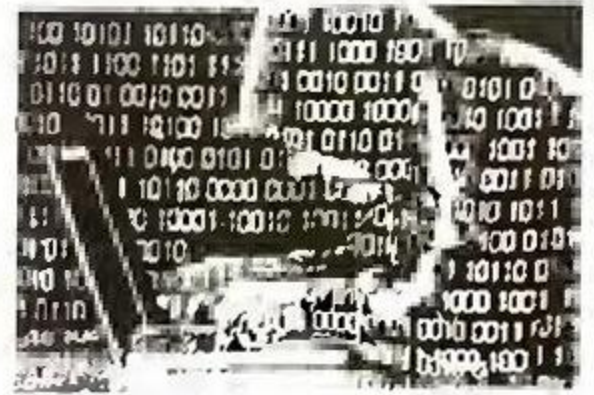


Fig. 11.7(a): Hacking

Thus, they help in improving the security of a software. An ethical hacker may exploit a website in order to discover its security loopholes or vulnerabilities. The findings are then reported to the website owner. Thus, ethical hacking is actually preparing the owner against any cyber attack.

A non-ethical hacker is one who tries to gain unauthorised access to computers or networks in order to steal sensitive data with the intent to damage or bring down systems. They are called black hat hackers or crackers. Their primary focus is on security cracking and data stealing. They use their skills for illegal or malicious purposes. Such hackers try to break through system securities for identity theft, monetary gain, to bring a competitor or rival site down, to leak sensitive information, etc.

An eavesdropping attack occurs when a hacker intercepts, deletes or modifies data that is transmitted between two devices. Eavesdropping, also known as sniffing or snooping, relies on unsecured network communications to access data in transit between devices. It typically occurs when a user connects to a network in which traffic is not secured or encrypted.

11.11.2 Phishing/Fraud Emails

Phishing is an attempt to acquire sensitive information such as usernames, passwords and credit card details (and sometimes, indirectly, money) by masquerading as a trustworthy entity in an electronic communication.

Phishing is an unlawful activity where fake websites or emails that look original or authentic are presented to the user to fraudulently collect sensitive and personal details, particularly usernames, passwords, banking and credit card details.



Fig. 11.7(b): Phishing

Phishing is typically carried out by email spoofing or instant messaging and it often directs the users to enter personal information on a fake website, the look and feel of which is identical to the legitimate one, the only difference being the URL of the website.

Communications purporting to be from social websites, email sites, peer-to-peer processors or IT administrators are often used to lure victims. Phishing emails contain links to websites that distribute malware.

The protective measures to be followed against Phishing include:

1. Never open or download a file from an unsolicited email, even from someone you know. You can call or email the person to double check that it really came from them.
2. Keep your operating system updated.
3. Use a reputable antivirus program.
4. Enable two-factor authentication whenever available.
5. Confirm the authenticity of a website prior to entering login credentials by looking for a reputable security trust mark.
6. Look for HTTPS in the address bar when you enter any sensitive personal information on a website to make sure your data will be encrypted.

11.11.3 Ransomware

Ransomware is malicious software that infects your computer and displays messages demanding a fee to be paid in order to access your system. This class of malware is a criminal moneymaking scheme that can be installed through deceptive links in an email message, instant message or website. It has the ability to lock a computer screen or encrypt important, predetermined files with a password.



Fig. 11.7(c): Ransomware

It uses scare tactics or intimidation to trick victims into paying up. It can come in the form of fake antivirus software in which a message suddenly appears claiming your computer has various issues and an online payment is necessary to fix them!

11.11.4 Illegal Downloading

Illegal downloading means obtaining files from the internet that you don't have the right to use. Illegal downloading is the root cause of digital piracy.

Digital piracy involves illegally sharing copyrighted media such as games, music, movies, TV shows and software. It does not matter whether you upload the content to share with others without permission or whether you download it for free or at a bargain price. Anyone involved in this type of operation is treated as law-breaker.



Fig. 11.7(d): Illegal Downloading

For example, if you are a member of a website that shares music files without permission from the recording labels, you are considered as involved in piracy, even if you have paid a fee to the website for downloading files. Another example could be if you are able to access and download movies or TV shows for free or at extremely low prices, which are usually illegal copies, and thus come within the ambit of piracy. To avoid downloading media illegally, obtain it from reputed dealers that have the rights to sell them.

Therefore, illegal downloading should be discouraged as it incurs heavy losses to developers/producers and the person who is carrying out this illegal downloading can face punishment or fine due to this unethical act.

11.11.5 Child Pornography

Child pornography is publishing and transmitting obscene material about children in electronic form. In recent years, child pornography has increased due to easy internet access and easily available videos on the internet. Child pornography is the most heinous crime which occurs and has led to various other crimes such as sex tourism, sexual abuse of the child, etc.

There are provisions under the cyber law to stop child pornography, leading to punishment like,

- 1st conviction with 2 to 5 years of imprisonment and a fine of ₹ 1 lakh.
- 2nd or subsequent conviction with an imprisonment of up to 7-10 years and a fine which may extend up to ₹ 10 lakh.

Child pornography laws provide severe penalties for producers and distributors in almost all Western societies, usually including incarceration, with shorter duration of sentences for non-commercial distribution depending on the extent and content of the material distributed. Convictions for possessing child pornography material also usually include prison sentences, but those sentences are often converted to probation for first-time offenders.

11.11.6 Cyber Scams and Frauds

The term 'Internet fraud' generally refers to any scheme that uses one or more components of the internet, such as chat rooms, email, message boards or websites, to present fraudulent solicitations to prospective victims, to conduct fraudulent transactions, or to transmit the proceeds of fraud to financial institutions or to others connected with the scheme.

The various cyber frauds that are carried out online across the world are:

- **Credit/Debit Card Fraud**—This involves the use of credit/debit card to obtain money or acquire properties without appropriate authorization. Fraudsters obtain credit/debit card numbers of victims from unsecured web media and use them to carry out illegal or unethical transactions.
- **Non-delivery of Goods/Service Fraud**—This refers to a scam in which people are encouraged to pay for goods and/or services via a web portal, and then nothing is delivered to the buyer.
- **Spoofing/Phishing Scam**—Spoofing is a type of fraud in which a fraudster masquerades as another person by using the other person's identity to transact business and obtain vital information such as bank account numbers, credit card numbers and associated passwords. Phishing is a form of spoofing in which the web page of a particular entity can be duplicated and positioned with URL for the purpose of luring people to divulge vital financial information.



Fig. 11.7(e): Internet Frauds



- **Identity Theft**—This entails the use of another person's personal information, without appropriate consent, for the purpose of fraudulent practices. Often personal information may even be leaked online or stolen during web surfing.

Identity theft is a type of fraud that involves using someone else's identity to steal money or gain other benefits. Online identity theft refers to an act of stealing someone else's personal information such as name, login details, etc., and then posing as that person online.

- **Auction Fraud**—In auction frauds, people are encouraged to participate in online auction and when money has been paid for specific items, the fraudster would send either a lower standard item or a counterfeit.

11.11.7 Preventing Cybercrime

Anyone using the internet should exercise some basic precautions. Here are some tips you can use to help protect yourself from cybercrime:

1. Keep the computer system up-to-date and take regular backup of important data.
2. Protect your personal information by managing your social media privacy settings.
3. Update your antivirus software on a regular basis.
4. Choose a strong password (combination of letter, number, symbols and special characters) and change it periodically. Do not use same password for all websites.
5. Before downloading any software, check the website address with HTTPS or lock symbol.
6. While using someone else's computer, don't allow the browser to save password or auto fill data, and try to browse in your private browser window activity.
7. For an unknown site, do not agree to use cookies when asked for, through a Yes/No option.
8. Perform online transaction like shopping, ticketing and other such services only through well-known and secure sites.
9. Always secure wireless network at home with strong password and regularly change it.
10. Do not download song, videos or software from untrusted websites. Often these files come with hidden malware such as ransomware or Trojan.
11. Never download attachments from emails that appear suspicious.
12. Adjust the settings in the web-browser. It may limit some functionality but can provide best protection from malicious content.



MEMORY BYTES

- Cyber law is a term used to describe legal issues related to the use of communications technology, particularly "cyberspace", i.e., the internet.
- Intellectual property refers to creation of the intellect (hence, the name); inventions, literary and artistic works, symbols, names, images and designs used in commerce are part of it.
- Digital Rights Management (DRM) is a term used for access control technologies that are used by hardware manufacturers, publishers, copyright holders and individuals to limit the use of digital content and devices.
- All software are protected under copyright. Copyright is owned by the developer or the publisher.
- Licences provide rules and guidelines for others to use your work. Open source licences help others to contribute to your work or project without seeking special individual permission to do so.
- GPL licensed code cannot be used for paid and proprietary software.

- All electronic waste is made up of deadly chemicals such as lead, cadmium, beryllium, mercury and brominated flame retardants.
- Identity theft occurs when someone, without your knowledge, acquires a piece of your personal information and uses it to commit a fraud.
- Phishing is an attempt to acquire sensitive information such as username, password, credit card details, etc.
- While entering any sensitive personal information on a website, look for HTTPS in the address bar to make sure your data is encrypted.
- An identity verification service is used to ensure that users or customers provide information that is associated with the identity of a real person.
- You should set up privacy settings for a social media site on your own.
- Proper ICT training should be imparted to teachers to solve the problem of gender bias in computer classroom teaching and learning.
- Certain software are made available for free public access. Free and Open Source Software (FOSS) allow users to not only access but also to modify (or improve) them.
- Cybercrimes include various criminal activities carried out to steal data or to break down important services. These include hacking, spreading viruses or malware, sending phishing or fraudulent emails, ransomware, etc.

OBJECTIVE TYPE QUESTIONS

1. Fill in the blanks.

- (a) Any information about you or created by you that exists in digital form is referred to as
- (b) Stealing someone's intellectual work and representing it as one's own is known as
- (c) Creative creations of the mind such as patents, trademark and copyright are property.
- (d) Any fraudulent business practice that extracts money from an ignorant person is called a(n)
- (e) Software which usually limit the functionality after a trial period are known as
- (f) Online is a theft of personal information in order to commit a fraud.
- (g) is defined as a crime in which computer is the medium of crime (hacking, phishing, spamming) or the computer is used as a tool to commit crimes.
- (h) The practice of attempting to acquire sensitive information from individuals over the internet by means of deception is called
- (i) means using another person's work, without permission or without paying for it, if it is being sold.

2. State whether the following statements are True or False.

- (a) Public Domain Software is free and can be used with restrictions.
- (b) An important factor of privacy is consumer consent.
- (c) Intellectual Property Rights are the rights of owners to decide how much information/data is to be shared or exchanged.
- (d) Copying and pasting data from internet or other digital resources is ethical.
- (e) E-document becomes legal after it is digitally signed.
- (f) The data stored in biometric database can be updated.
- (g) Source code of proprietary software is normally available.
- (h) Free software is same as freeware.
- (i) Never respond to an email or advertisement that claims you have won something.

3. Multiple Choice Questions (MCQs)

- (a) Using someone else's Twitter handle to post something will be termed as:
 - (i) Fraud
 - (ii) Identity theft
 - (iii) Online stealing
 - (iv) Violation

- (b) Standard security protocol that establishes encrypted links between a web server and a browser is called
- | | |
|---------------------------------|--------------------------------------|
| (i) Online safety technology | (iii) SDT technology |
| (iii) Web encryption technology | (iv) Secure Sockets Layer Technology |
- (c) Intellectual Property Rights protect the use of information and ideas that are of:
- | | |
|--------------------|-----------------------|
| (i) Ethical Value | (iii) Moral Value |
| (iii) Social Value | (iv) Commercial Value |
- (d) Credit card fraud may include:
- | | |
|-----------------------------|--|
| (i) Stealing of credit card | (ii) Unauthorized and illegal use of credit card |
| (iii) Stealing information | (iv) Phishing |
- (e) The term "Intellectual Property Rights" covers:
- | | | | |
|----------------|-----------------|---------------|-------------------|
| (i) Copyrights | (ii) Trademarks | (iii) Patents | (iv) All of these |
|----------------|-----------------|---------------|-------------------|
- (f) Which of the following is an open-source software?
- | | | | |
|-----------------------|----------------------|-------------|--------------------|
| (i) Microsoft Windows | (ii) Adobe Photoshop | (iii) MySQL | (iv) MS PowerPoint |
|-----------------------|----------------------|-------------|--------------------|
- (g) Online personal account, personal websites are examples of:
- | | | | |
|--------------------|-----------------------|---------------------------|------------------------|
| (i) Digital Wallet | (ii) Digital Property | (iii) Digital Certificate | (iv) Digital Signature |
|--------------------|-----------------------|---------------------------|------------------------|
- (h) Unsolicited commercial email is known as
- | | | | |
|----------|--------------|-------------|--------------|
| (i) Spam | (ii) Malware | (iii) Virus | (iv) Spyware |
|----------|--------------|-------------|--------------|
- (i) Which of the following is not a type of a cybercrime?
- | | |
|----------------------|--|
| (i) Data theft | (ii) Forgery |
| (iii) Damage to data | (iv) Installing antivirus for protection |

SOLVED QUESTIONS

1. What do you understand by 'Privacy of data'?

Ans. The ethical and legal rights that individuals have with regard to control over discussions and use of their personal information is known as privacy of data.

2. If someone hacks your website, who would you complain to?

Ans. The complaint has to be lodged with the police under the IT Act.

3. What is the importance of cyber law?

Ans. Cyber law is a term used to describe the legal issues related to use of communications technology, particularly "cyberspace", i.e., the Internet. Some of the issues addressed by cyber law are cybercrime, electronic commerce, intellectual property, data protection and data privacy. Cyber laws protect people from online frauds and scams. These laws help prevent financial crimes and identity theft that happen online.

4. Write two applications of cyber law.

[CBSE Delhi 2015]

Ans. Cyber law encompasses a wide variety of political and legal issues related to the internet and other communication technologies, including intellectual property, privacy, freedom of expression and jurisdiction.

5. Malini has stolen a credit card. She uses this credit card to purchase a laptop. What type of offence has she committed?

Ans. She has committed a fraud.

6. Name the primary law in India dealing with cybercrime and electronic commerce.

Ans. The primary law to deal with cybercrime and electronic commerce issue is Information Technology Act, 2000.

7. Sujata received an email from her bank stating that there is a problem with her account. The email provides instructions and a link, by clicking on which she can log on to her account and fix the problem. Help Sujata by telling her the precautions she should take on receiving such types of emails.

Ans. She should check whether it is a valid bank site or not by checking in the url https. It is always better to type the url and then log on to the site. She should not click on the link provided in the email.

8. Explain any two ways in which technology can help students with disabilities.

Ans. Different types of ICT tools assist people with learning disabilities to achieve positive outcomes. They are:

- Talking Word processors
- Screen Readers
- Conversion of local language to Braille
- Eye Tracking mouse

9. Explain the role of online social media campaigns, crowd sourcing and smart mobs in society.

Ans. Role of Social Media Campaigns:

- A social media campaign should focus around a singular business goal, whether it is on Facebook or Instagram. Common goals for social media campaigns include:
 - Getting feedback from users.
 - Building email marketing lists
 - Increasing website traffic
- Crowd sourcing is the practice of engaging a 'crowd' or group for a common goal—often innovation, problem-solving or efficiency. It is powered by new technologies, social media and Web 2.0. Crowd sourcing can take place on many different levels and across various industries.
- Smart mobs, so named because each person in the group uses technology to receive information on where to go and what to do. The ability to stay on top of current events makes smart mobs extremely effective.

10. Which of the following come under cybercrime?

[HOTS]

1. Theft of a brand new sealed laptop.
2. Access to a bank account for an unauthorized money transaction.
3. Modification in a company's data with unauthorized access.
4. Photocopying a printed report.

Ans. (2) and (3)

11. List three points of network security components.

[HOTS]

Ans. The three network security components are:

- Anti-virus and anti-spyware.
- Firewall, to block unauthorized access to your network.
- Intrusion Prevention Systems (IPS) to identify fast-spreading threats such as zero-day or zero-hour attacks.

12. Name the crimes for which cyber laws are enforced strictly in India.

Ans. These are:

- | | |
|--------------------------------|--|
| (a) cybercrimes, | (b) electronic and digital signatures, |
| (c) intellectual property, and | (d) data protection and privacy. |

13. What are privacy laws?

Ans. Privacy laws refer to laws that deal with regulating, storing and using personally identifiable information of individuals, which can be collected by government, public or private organizations, or other individuals. Privacy laws are considered within the context of an individual's privacy rights or within reasonable expectations of privacy.

14. Explain phishing.

Ans. Phishing is the fraudulent attempt to obtain sensitive information such as usernames, passwords and credit card details (and money), often for malicious reasons, by disguising as a trustworthy entity in an electronic communication. Phishing is typically carried out by email spoofing or instant messaging, and it often directs users to enter personal information at a fake website, the look and feel of which is identical to the legitimate one and the only difference is the URL of the website in question.

15. List some of the risks that internet users face.

Ans. The main risks internet users face include:

- Inappropriate or unwanted contact (cyberbullying and spamming)
- Inappropriate or inaccurate content (pornography and hate sites)
- Deceptive or fraudulent commerce (counterfeit and malicious sites)

16. What does a phishing email look like?

[HOTS]

Ans. Typically, a phishing email appears to come from a financial institution, a large company, a chain store, a social networking site, or a government agency. The messages try to mimic a legitimate site by using the same or similar colours, logos, fonts and layout. And they often include a link to a legitimate-looking but phony web page that asks you to enter personal information. One tip-off that an email may be phishing is the use of phrases such as "Verify your account" and "Your account will be closed" if you don't provide certain sensitive information such as login name and password. A legitimate business will never ask for such personal information via email. Promising big lottery winnings, prizes or other types of windfall if you pay money upfront is another common phishing scam.

17. What if I have already responded to a phishing email and fear I may be a victim of identity theft? [HOTS]

Ans. Immediately change the password to your online account, and notify the fraud department of the institution that was mimicked. When you receive your credit reports, review them carefully for accounts you don't recognize, which may be signs of identity theft. Consider placing a security freeze on each report, which means that no new credit can be extended without your personal approval.

18. What are the different ways in which authentication of a person can be performed?

Ans. The different methods of user identification are:

1. **Password:** This is something that the user should know since they start their activity.
2. **Token:** This is something the user is provided with and should have it to perform an activity.
3. **Biometrics:** This means biological characteristics of a person registered for verification.
4. **OTP:** A one-time pin or password is sent to the user through which they verify their identity.

19. How can we prevent identity thefts?

Ans. Identity theft refers to the acquisition of personal data of the victim and using it for illegal purposes. It is the most common type of fraud that may lead to financial losses and at times may result in criminal action as the victim might be personified.

A few steps to follow in order to prevent identity thefts:

- Ensure a strong and unique password
- Avoid posting confidential information online
- Do not post personal information on social media
- Shop from known and trusted websites only
- Use the latest version of the browsers
- Install advanced malware and spyware tools
- Use specialized security solutions against financial data
- Always update your system and the software
- Protect your social security number
- Download only well-known apps and share limited details

20. What do you understand by the term 'open data'?

Ans. Open data is the idea that some data should be freely available to everyone to use and republish as they wish, without restrictions of copyright, patents or other mechanisms of control. The goals of the open data movement are similar to those of other "open" movements such as open source, open hardware, open content, open education, open educational resources, open government, open knowledge, open access, open science, and open web.

21. "Open data can be a powerful force for public accountability." Explain.

Ans. "Open data can be a powerful force for public accountability—it can make existing information easier to analyze, process, and combine than ever before, allowing a new level of public scrutiny." Governments that enable public viewing of data can help citizens engage within the governmental sectors and "add value to that data."

22. What are the different types of cybercrimes?

Ans. Common types of cybercrimes include online bank information theft, identity theft, online predatory crimes and unauthorized computer access. More serious crimes like cyberterrorism are also of significant concern. Cybercrime encompasses a wide range of activities, but these can generally be broken into two categories:

- Crimes that target computer networks or devices. These types of crimes include viruses and denial-of-service (DoS) attacks.
- Crimes that use computer networks to advance other criminal activities. These types of crimes include cyberstalking, phishing and fraud or identity theft.

23. What does cybercrime mean?

Ans. Cybercrime is defined as a crime in which a computer is the object of the crime (hacking, phishing, spamming) or is used as a tool to commit an offence (child pornography, hate crimes, etc.). Cybercriminals may use computer technology to access personal information, business trade secrets, or use the internet for exploitative or malicious purposes.

24. What security measures should be taken against phishing attacks?

Ans. The following security precautions are recommended against phishing:

- Use updated computer security tools, such as anti-virus software, spyware and firewall.
- Never open unknown or suspicious email attachments.
- Never divulge personal information requested by email, such as your name or credit card number.
- Double check the website URL for legitimacy by typing the actual address in your web browser.
- Verify the website's phone number before placing any calls to the phone number provided via email.

25. How is digital piracy related to illegal downloads?

Ans. Digital piracy involves illegally sharing copyrighted media such as games, music, movies, TV shows and software. It does not matter whether you upload the content to share with others without permission or whether you download it for free or for a bargain price.

Illegal downloads from the internet involve violation of copyright laws where users download material, such as music, movies, and other forms of media, without properly purchasing the product or doing so without proper permission of copyright holder. Hence, illegal downloads result in digital piracy.

26. Explain the term 'Web beacons'.

[HOTS]

Ans. Pictures in email messages—also called 'Web beacons'—can be adapted to secretly send a message back to the sender. Spammers rely on information returned by these images to locate active email addresses. Images can also contain harmful code and be used to deliver a spammer's message in spite of filters.

27. How can we block Web beacons?

[HOTS]

Ans. (a) The best defence against Web beacons is to prevent pictures from downloading until you've had a chance to review the message.

(b) Beware of malware disguised as holiday greetings in email.

(c) Approach links in email, on social networking sites, or in IMs with caution.

28. Jagdish was asked to collect information on "Digital India". He surfed the net and downloaded a document. He submitted the same documents to his teacher. Do you think it is right? If not, why?

Ans. No, this is wrong. It is a cyber crime and IPR infringement.

UNSOLVED QUESTIONS

1. What do you mean by cybercrime?
2. What is meant by cyber ethics? How are they important to us?
3. What do you mean by identity theft? How can we prevent it?
4. What do you understand by illegal download?
5. What is child pornography?
6. What is phishing?
7. How do people unknowingly commit cybercrime?
8. What are the different ways to do phishing?
9. What precautions should be taken to avoid phishing?
10. What procedure is followed by the police to track/investigate cybercrime cases?
11. What do you understand by the concept of cyber forensics?
12. What precautions should be taken so that students do not indulge in cybercrime unknowingly?
13. What is meant by gender discrimination? How does it hamper classroom teaching and learning process?
14. Differentiate between Proprietary Licence and Open Source Licence.

15. How secure is Biometrics Authentication System?
16. What is the relationship between ethics and law?
17. What is the difference between fairness and ethics in law?
18. How do ethical norms differ from the law?
19. What do you mean by privacy law?
20. What is the difference between threat, vulnerability and risk?
21. What is the goal of information security within an organization?
22. In how many ways can you secure your system from cyberattacks?
23. Write the advantages of Biometrics in school.
24. Differentiate between Hacker and Cracker.
25. What do you mean by illegal downloads? What are its repercussions?
26. What do you mean by virus?
27. What do you mean by Spyware and Adware?
28. What do you mean by Malware?
29. What do you mean by Worm?
30. What do you mean by Trojan Horse?
31. What is the difference between Worm and Trojan Horse?
32. What is the difference between Malware and Worm?
33. How does technology impact the society? Explain with the help of an example.
34. Write the drawback of social media networks.
35. Write the names of five social media sites which are very popular.
36. What is plagiarism?
37. Match the following:

Column A

Plagiarism

Hacking

Credit card fraud

Digital footprint

Column B

Fakers, by offering special rewards or money prize asked for personal information, such as bank account information

Copy and paste information from the internet into your report and then organize it.

The trail that is created when a person uses the internet.

Breaking into computers to read private emails and other files.

38. Write the differences between the following:
 - (a) Copyrights and Patents
 - (b) Plagiarism and Copyright infringement
 - (c) Non-ethical hacking and Ethical hacking
 - (d) Active and Passive footprints
 - (e) Free software and Free and open-source software
39. Cite examples depicting that you were a victim of following cybercrime. Also, cite provisions in IT Act to deal with such a cybercrime.
 - (a) Identity theft
 - (b) Credit card account theft
40. Name the different types of intellectual property.
41. Ravi received a mail from IRS department. On clicking "Click here", he was taken to an official-looking website, like IRS.gov. He uploaded some important information on it. Identify and explain the cybercrime discussed in this scenario.
42. Shreya was not able to attend the class and she called up Rajni enquiring about the topic discussed in the class. Rajni informed her that tuples in Python was discussed and also downloaded a PowerPoint presentation from the Internet. She mailed the presentation to her after changing the name to "made by Rajni" saying she had made this presentation for her.

Identify and explain the cybercrime being discussed in the above example.
43. What do you understand by trademark and intellectual property?
44. What is data privacy? Websites that you visit collect what type of information about you?

45. What do you mean by confidentiality of information?
46. What is the role of firewall?
47. What do you mean by cybercrime?
48. Name some common social networking sites.
49. Define Digital footprint.

CASE-BASED/SOURCE-BASED INTEGRATED QUESTIONS

1. Ahaana is down with fever. She decides not to go to school the next day. Next evening she calls up her classmate, Sabhya, and enquires about the Computer class. She also requests him to explain the concept. Sabhya says, "Mam taught us how to use dictionary in Python". Further, he politely says, "Give me some time, I will email you the material which will help you understand dictionary in Python." Sabhya quickly downloads a 2-minute clip from the internet explaining the concept of dictionary in Python. Using video editor, he adds the text "Prepared by Sabhya" in the downloaded video clip. He then emails the modified video clip to Ahaana. This act of Sabhya is an example of:
 - (a) Fair use
 - (b) Hacking
 - (c) Copyright infringement
 - (d) Cybercrime

Ans. This is an act of copyright infringement. Copyright infringement occurs when we use another person's work without either obtaining their permission or without paying for it, if it is being sold. This is what Sabhya has done and, in turn, violated the copyright laws.

2. After a fight with your friend, you did the following activities. Which of these activities is not an example of cyberbullying?
 - (a) You sent an email to your friend with a message saying, "I am sorry".
 - (b) You sent a threatening message to your friend saying, "Do not try to call or talk to me".
 - (c) You created an embarrassing picture of your friend and uploaded on your account on a social networking site.

Ans. You sent an email to your friend with a message saying, "I am sorry". This activity is not counted under cyberbullying.

3. Ravi received an email from First Generic Bank (as shown below). On clicking the link, he was taken to a site designed to imitate an official looking website. He uploaded some important information on it. Identify and explain the cybercrime being discussed in the following scenario:

From: First Generic Bank <accounts@firstgenericbank.com>
 Subject: Please update your account information
 Date: Sep 12, 2018 3:23 PM PST

Dear First Generic Bank user,

As a courtesy to our valued customers, First Generic Bank conducts regular account information verification processes. During the most recent process, we found that we could not verify your information.

In order to ensure your account information is not made vulnerable, please visit <http://www.firstgenericbank.com/account-updateinfo.com>.

Please click on the above link to our Website and confirm or update your account information. If you do not do this within 48 hours of receipt of this e-mail, you will not be able to use your First Generic Bank account for 30 days. This is an extra precaution we take to ensure your account remains secure.

Sincerely,
 First Generic Bank

Ans. The cybercrime discussed above is Phishing. Spoofing is a type of fraud in which a fraudster masquerades as another person by using the other person's identity to transact business and obtain vital information such as bank account numbers, credit card numbers and associated passwords. Phishing is a form of spoofing in which the web page of a particular entity can be duplicated and positioned with URL for the purpose of luring people to divulge vital financial information. Thus, in the above case, Phishing is the cybercrime committed.